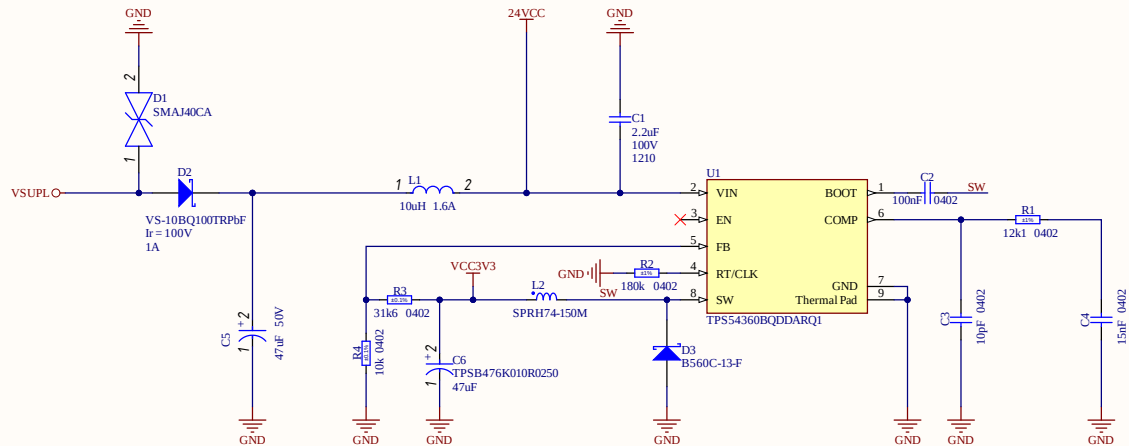




Изм.	Лист	№ докум.	Подп.	Дата
Разраб.		Дунаевский		
Пров.		Дунаевский		
Т.контр.				
Нач.сект.				
Н.контр.				
Утв.		Глазьяко		

ETH_MODULE_8AOC ЭЭ

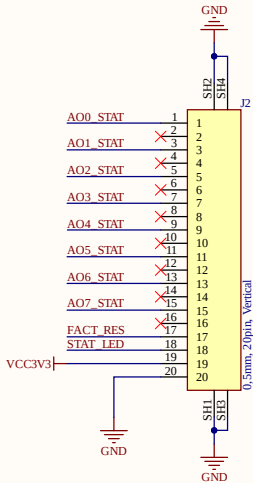
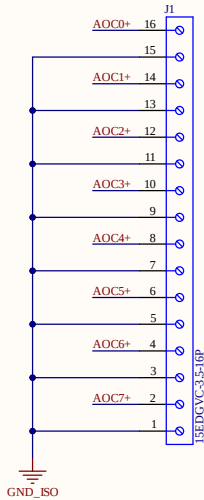
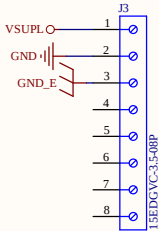
Модуль вывода аналог.
сигналов 8АО 4-20mA

СХЕМА ЭЛЕКТРИЧЕСКАЯ
ПРИНЦИПИАЛЬНАЯ

Лист	1	Листов	5
FATEC			

Инв.№ подл.	Подп.и дата	Взам.инв.№	Инв.№ дубл.	Подп. и дата

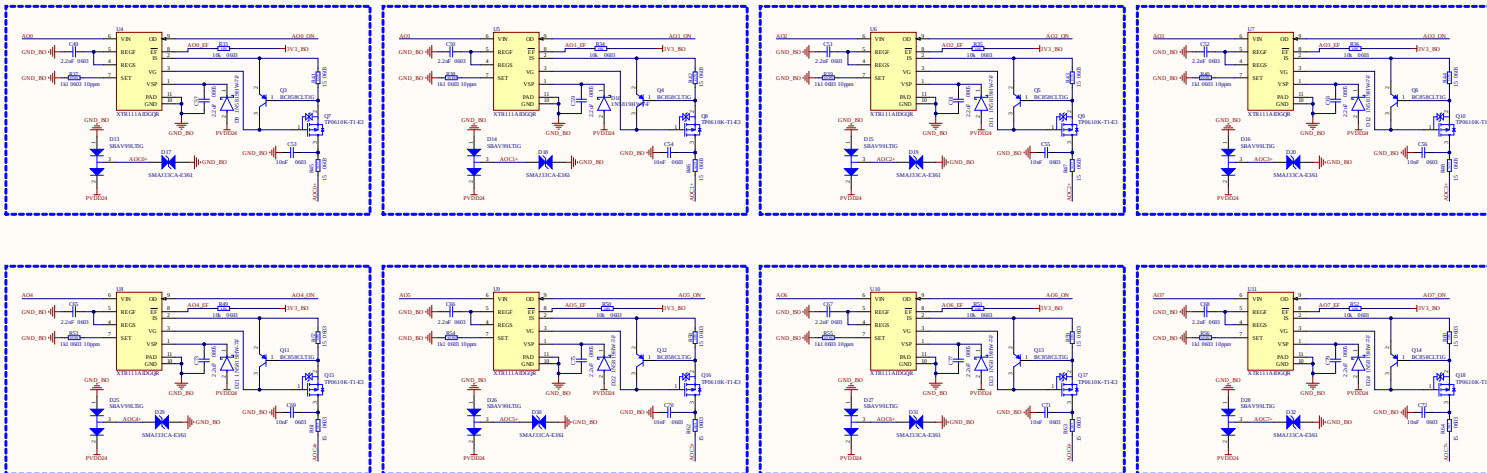
ETH_MODULE_8AOC ЭЭ



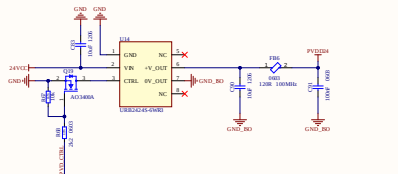
Изм.	Лист	№ докум.	Подп.	Дата

ETH_MODULE_8AOC ЭЭ

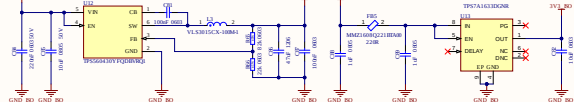
Hight-side, 0V, 2.5V - 0.22mA



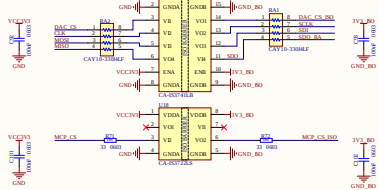
ISO 24V power supply



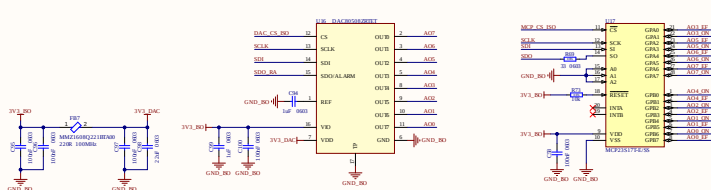
ISO 3V3

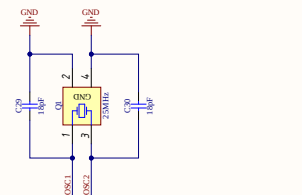
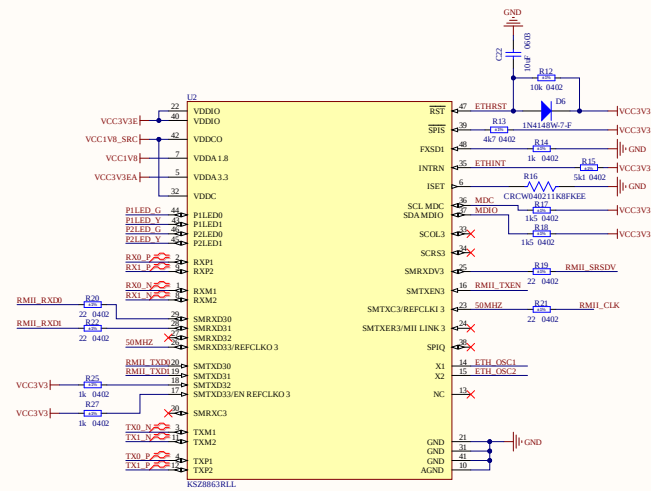
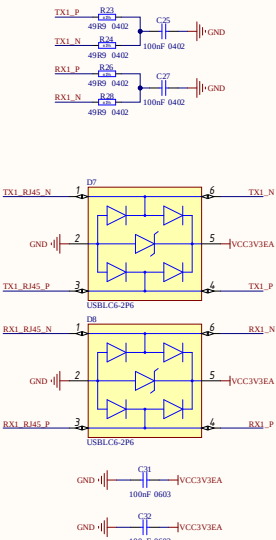
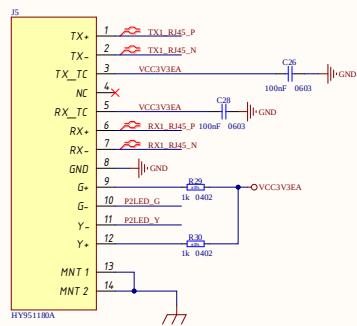
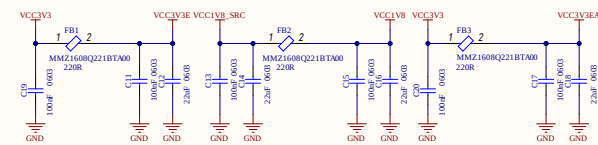
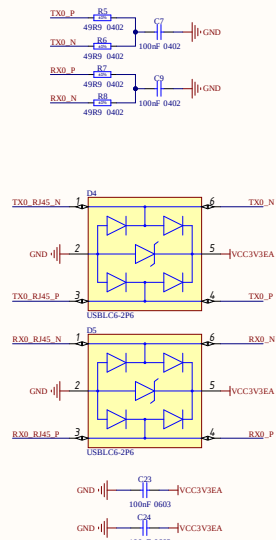
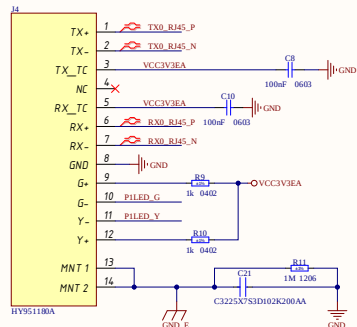


Isolators



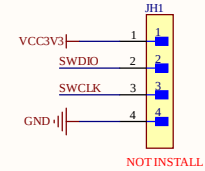
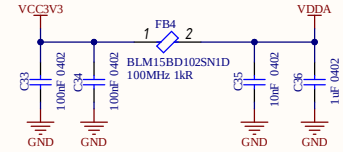
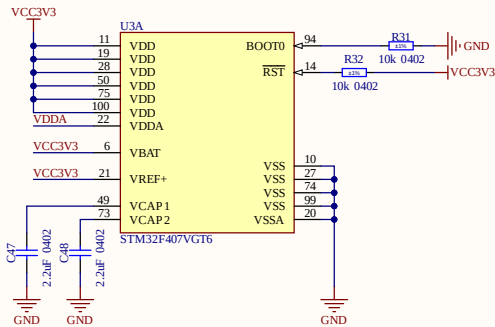
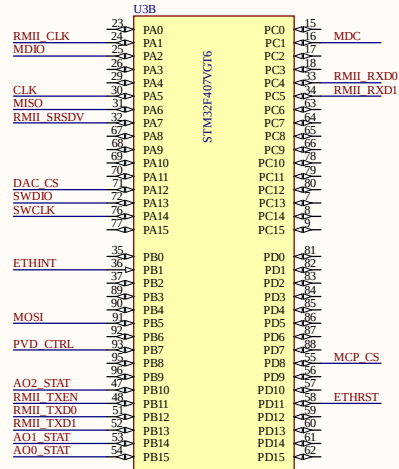
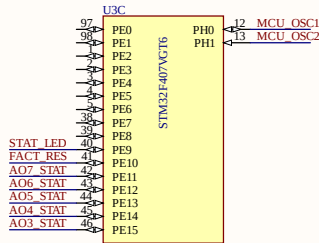
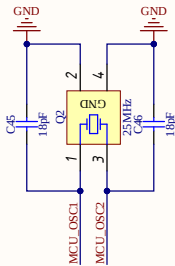
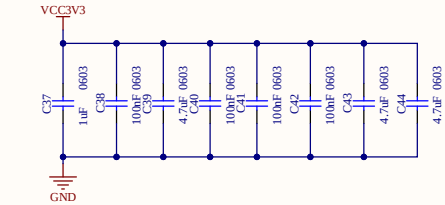
DAC





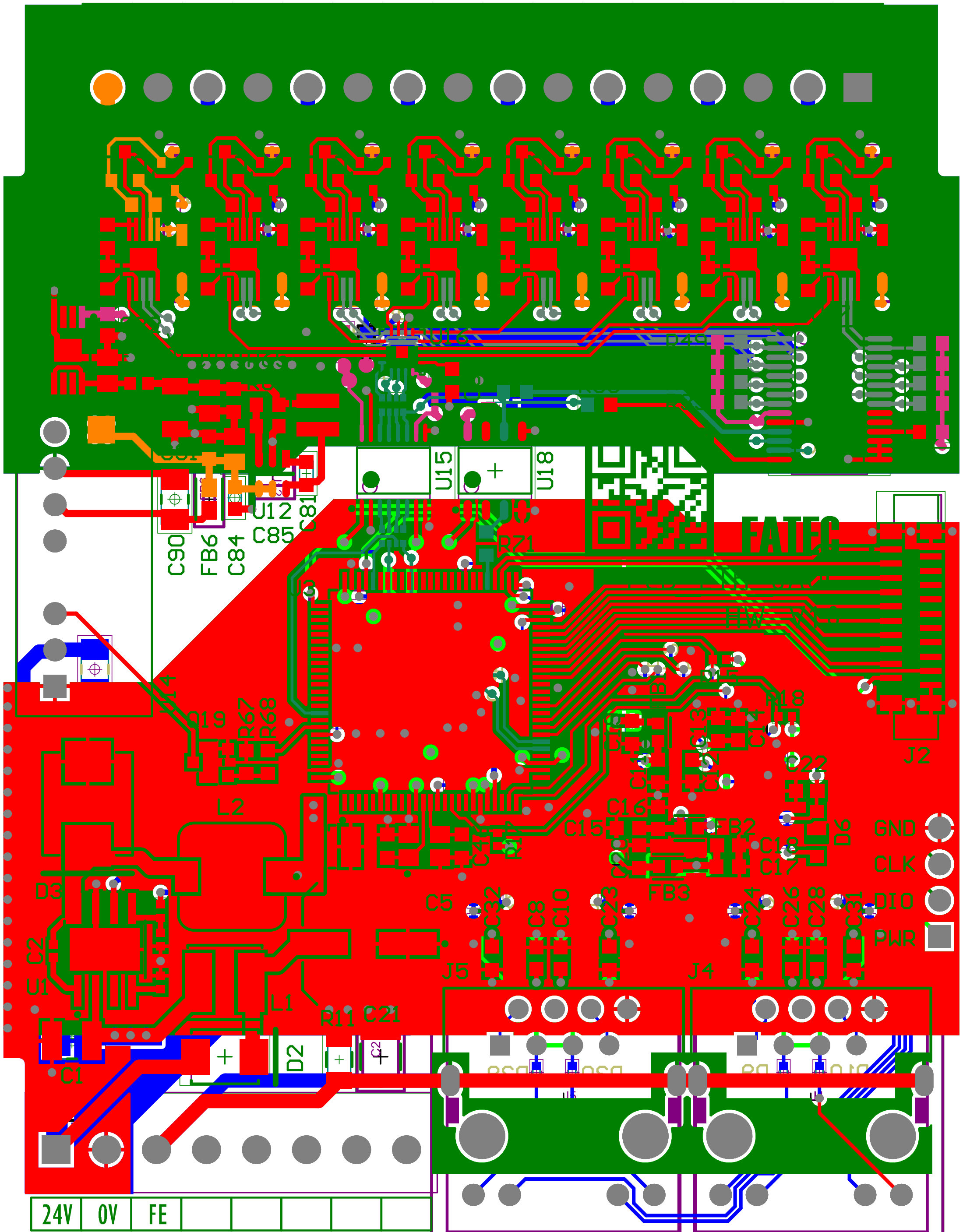
Инв.№ подл.	Подп.и дата	Взам.инв.№	Инв.№ дубл.	Подп. и дата

ETH_MODULE_8A0C ЭЗ

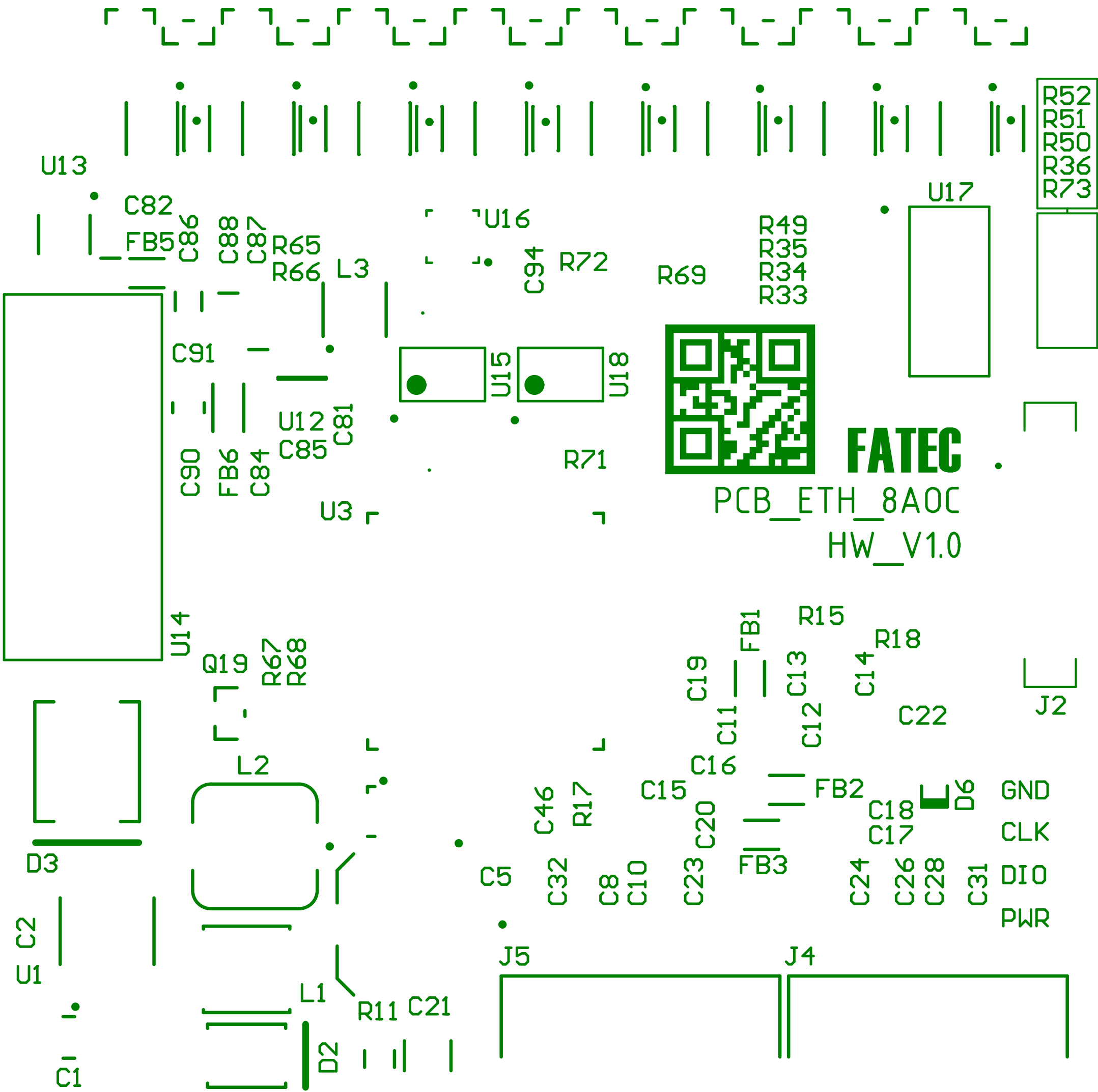


Изм.	Лист	№ докум.	Подп.	Дата

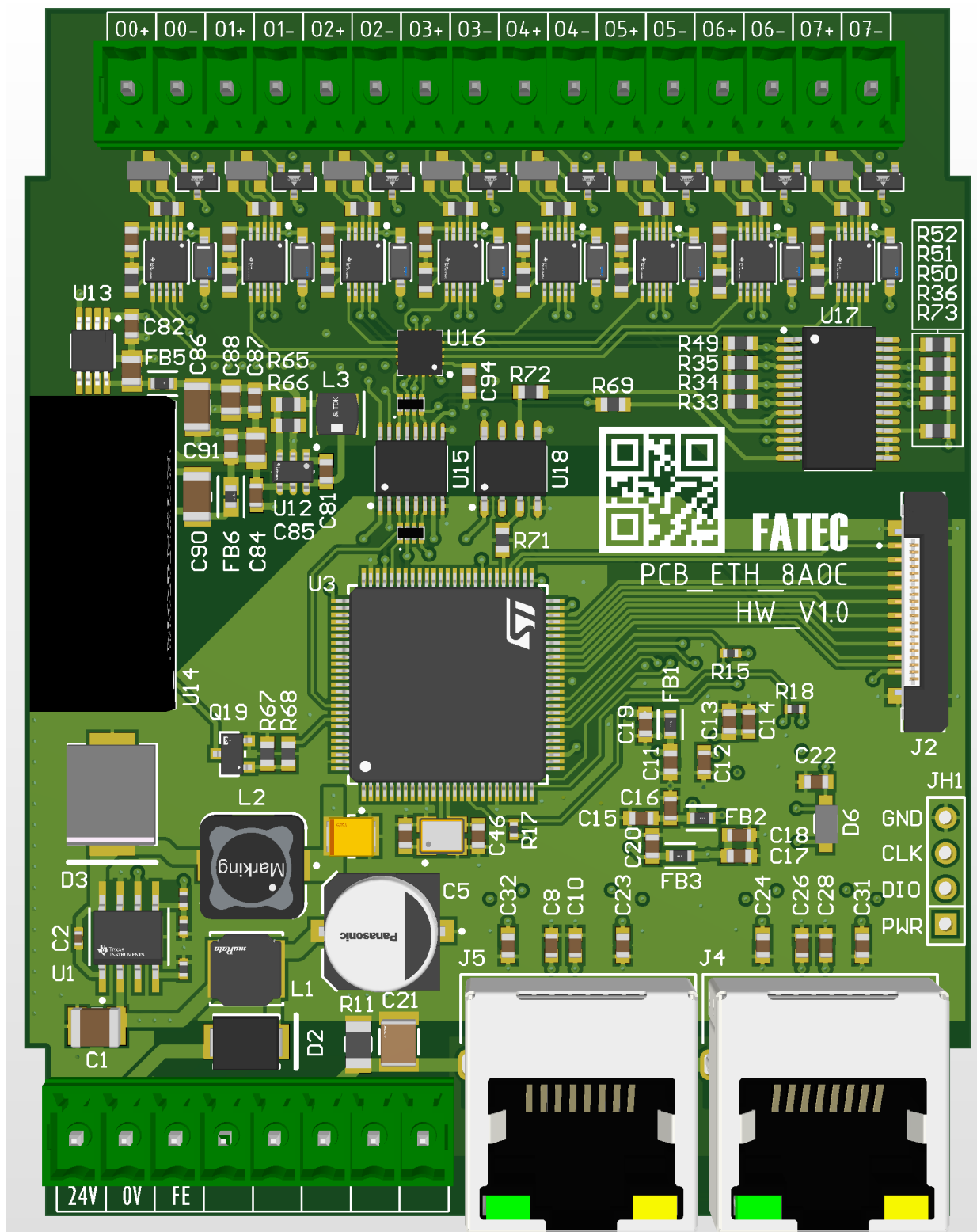
ETH_MODULE_8A0C ЭЗ

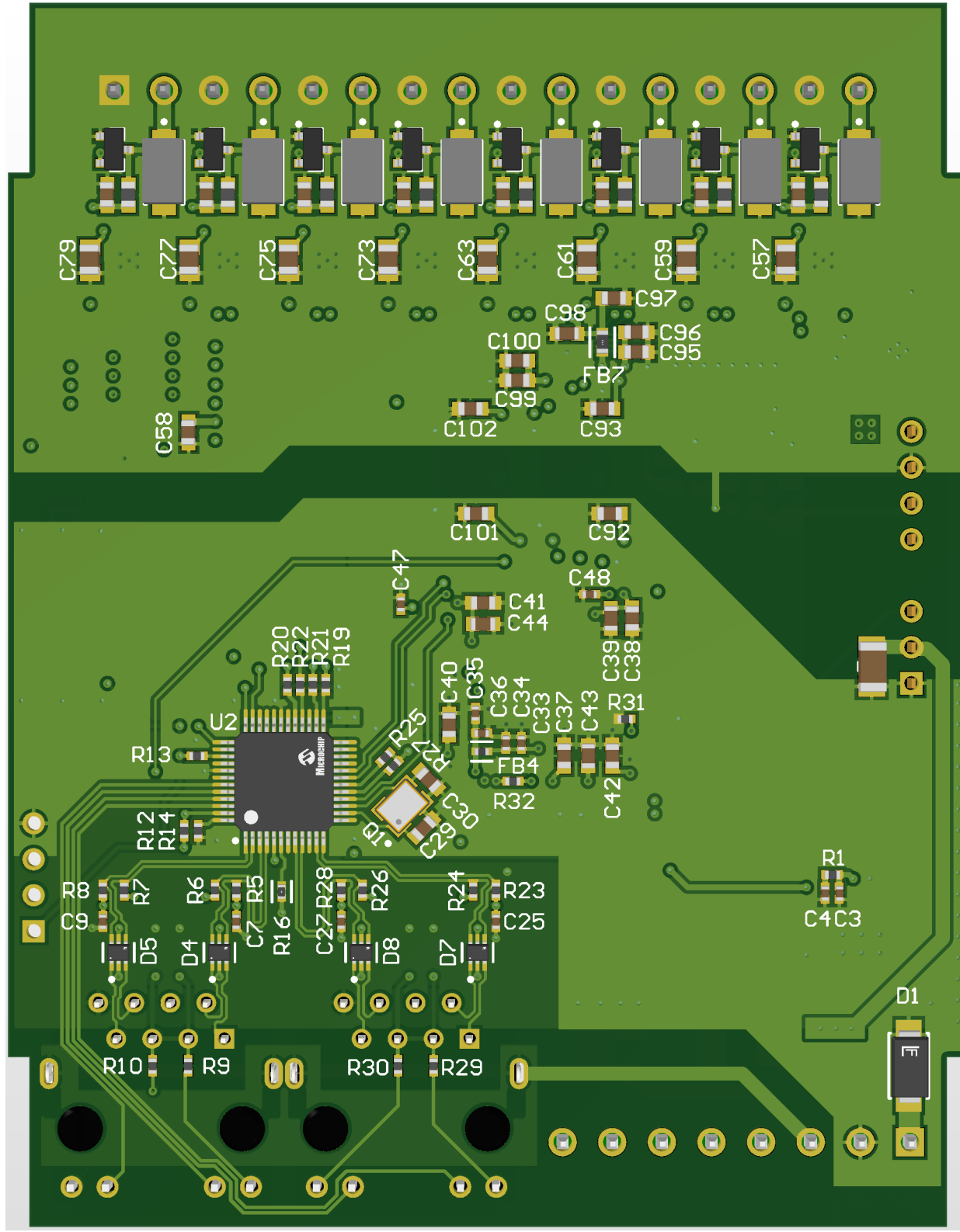


00+	00-	01+	01-	02+	02-	03+	03-	04+	04-	05+	05-	06+	06-	07+	07-
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

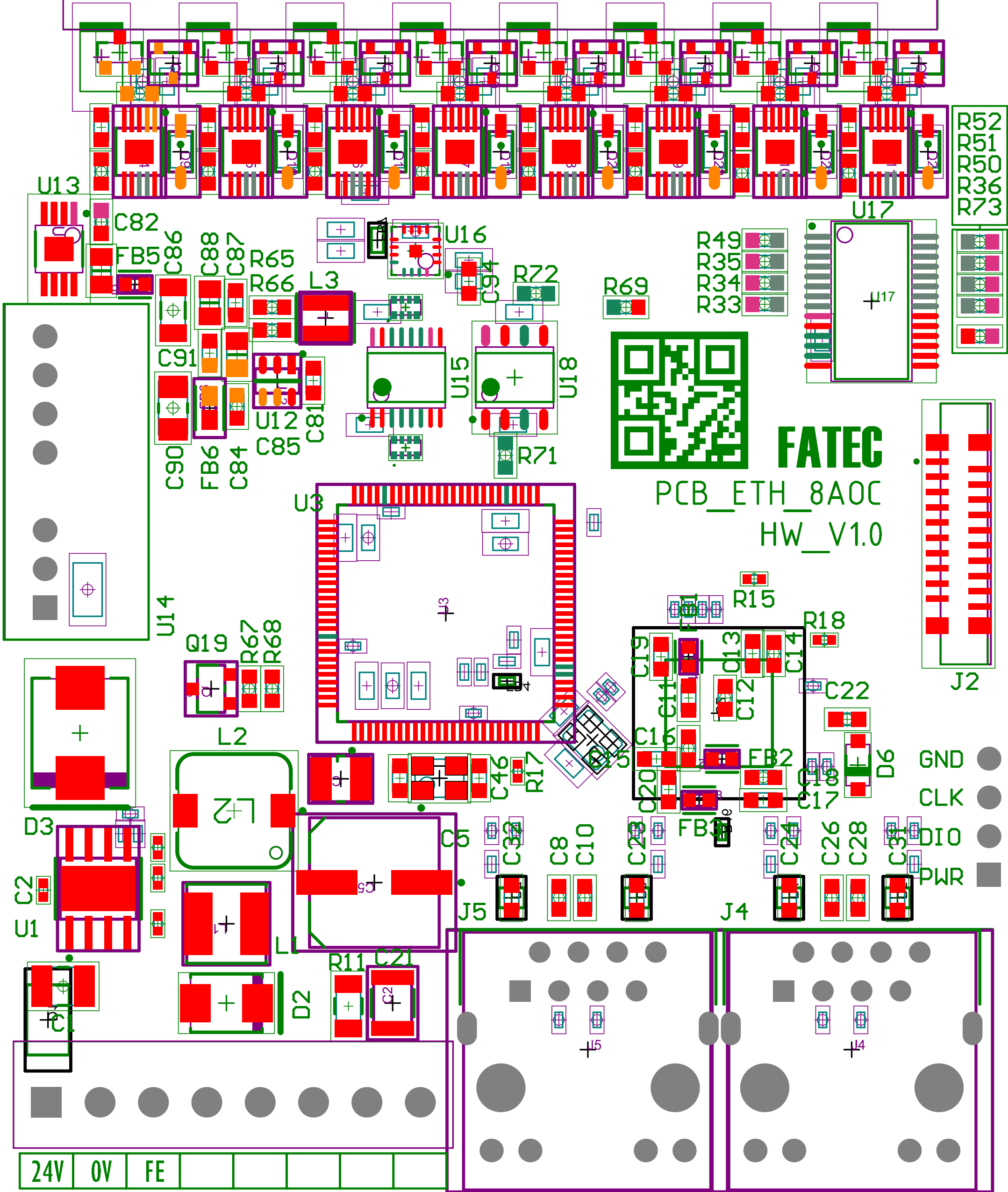


24V	0V	FE					
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00+ 00- 01+ 01- 02+ 02- 03+ 03- 04+ 04- 05+ 05- 06+ 06- 07+ 07-



FATEC

PCB_ETH_8A0C

HW_V1.0

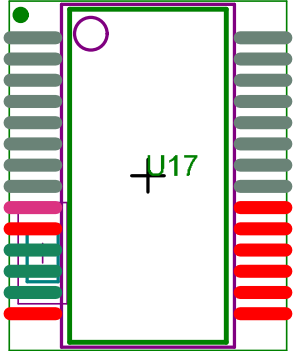
R52
R51
R50
R36
R73



R49
R35
R34
R33



U17



J2

GND
CLK
DIO
PWR

D6

C18

C17

FB2

C16

C15

C14

C13

C12

C11

C10

C9

C8

C7

C6

C5

C4

C3

C2

C1

U1

D3

J4

C28

C26

C24

FB3

C23

C20

C19

C18

C17

C16

C15

C14

C13

C12

C11

C10

C9

C8

C7

C6

C5

C4

J5

C46

R17

C45

C44

C43

C42

C41

C40

C39

C38

C37

C36

C35

C34

C33

C32

C31

C30

C29

C28

C27

C26

J2

C22

R18

C21

R15

C20

C19

C18

C17

C16

C15

C14

C13

C12

C11

C10

C9

C8

C7

C6

C5

C4

C3

J4

C28

C26

C24

FB3

C23

C20

C19

C18

C17

C16

C15

C14

C13

C12

C11

C10

C9

C8

C7

C6

C5

C4

J5

C46

R17

C45

C44

C43

C42

C41

C40

C39

C38

C37

C36

C35

C34

C33

C32

C31

C30

C29

C28

C27

C26

J2

C22

R18

C21

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C15

C14

C13

C12

C11

C10

C9

C8

C7

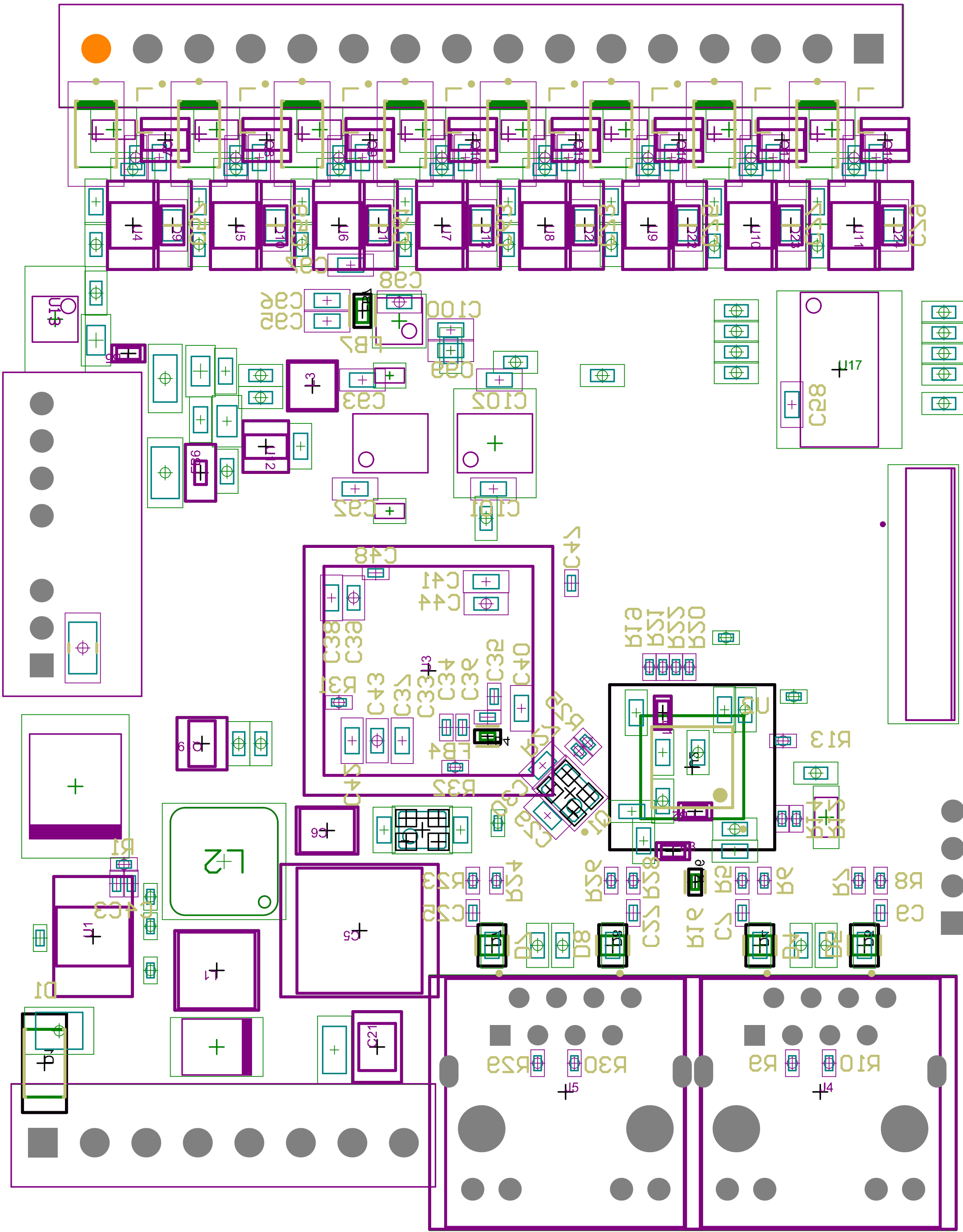
C6

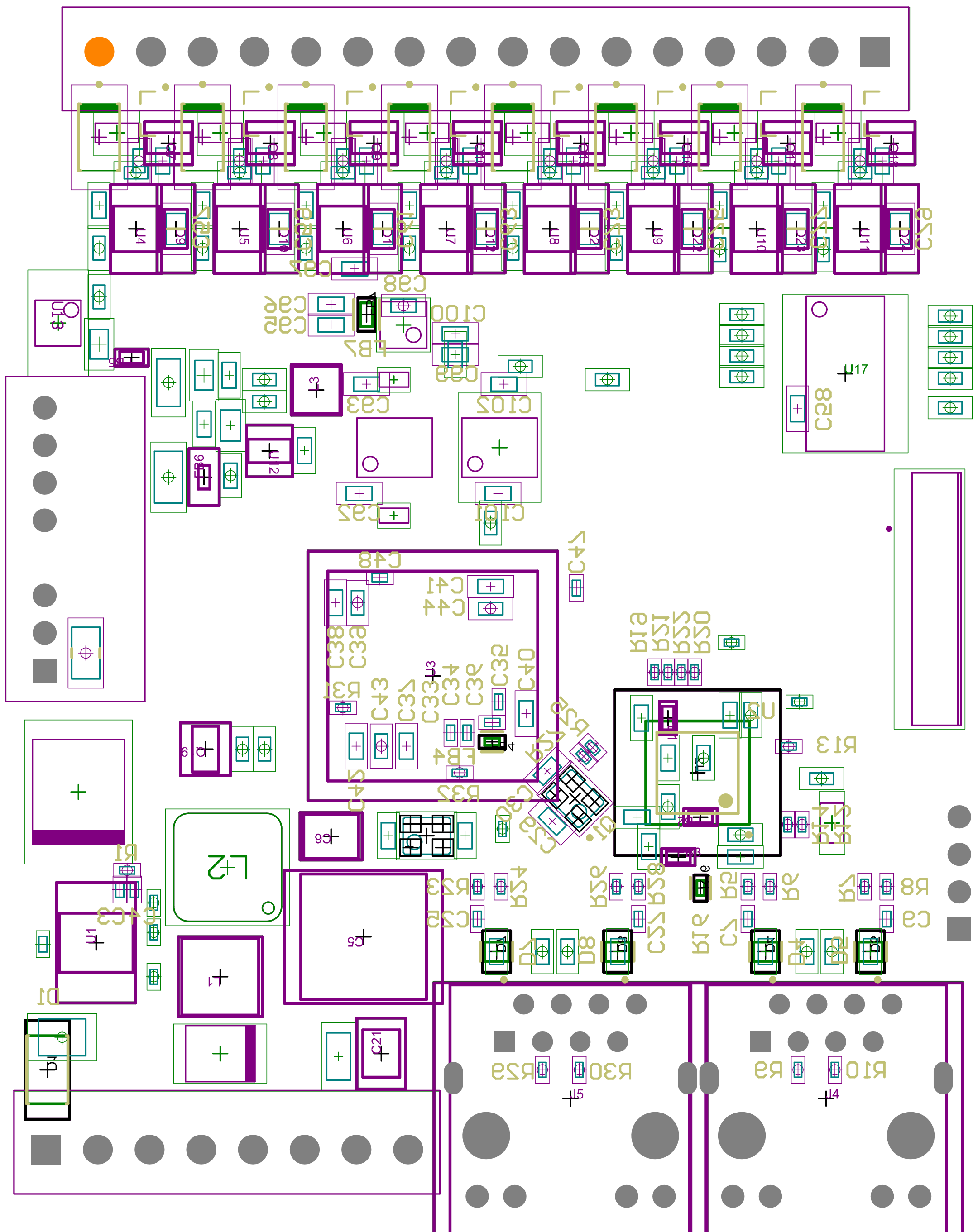
C5

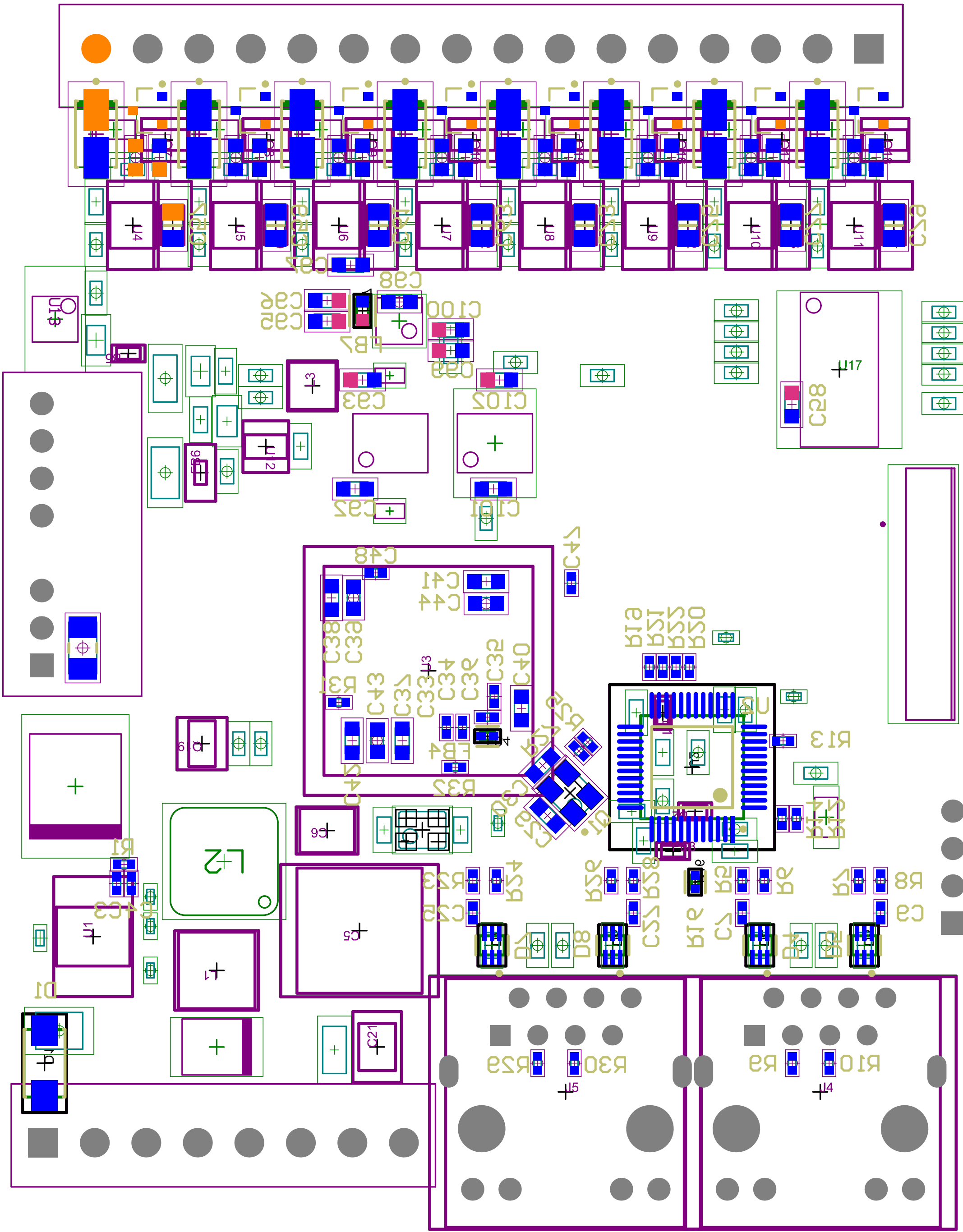
C4

C3

24V 0V FE







Bill of Materials					
Bill of Materials for Project [PLCJS_ETH_MODULE_8AOC_D4MG_STM32F407VGT6]					
Source Data From: PLCJS_ETH_MODULE_8AOC_D4MG_STM32F407VGT6.PrjPcb					
Project: PLCJS_ETH_MODULE_8AOC_D4MG_STM32F407VGT6.PrjPcb					
Variant: None					
Creation Date: 22.09.2026 11:27					
Print Date: 46287 46287,47718					
Footprint	Comment	LibRef	Designator	Description	Quantity
CAPC3225X2	2.2uF	CC1210_2.2UF	C1, C47, C48, C57, C59, C61, C63, C73,	Non-polarised capacitor	11
CAPC1005X5	100nF	CC0402_100NF	C2, C7, C8, C9, C10, C11, C13, C15, C17,	Non-polarised capacitor	37
CAPC1005X5	10pF	CC0402_10PF	C3	Non-polarised capacitor	1
CAPC1005X5	15nF	CC0402_15NF	C4	Non-polarised capacitor	1
FP-FKE-MFG	EEEEFK1H470P	CMP-05427-	C5	CAP ALUM 47UF 20% 50V SMD	1
FP-TPSB-	TPSB476K010R	CMP-04424-	C6	CAP TANT 47UF 10% 10V 1411	1
CAPC1608X9	22uF	CC0603_22UF	C12, C14, C16, C18, C98	Non-polarised capacitor	5
FP-C3225-	C3225X7S3D10	CMP-08246-	C21	Multilayer Ceramic Capacitors 1nF ±10%	1
CAPC1608X8	10uF	CC0603_10UF	C22, C82, C83, C85, C90	Non-polarised capacitor	5
CAPC1608X8	18pF	CC0603_18PF	C29, C30, C45, C46	Non-polarised capacitor	4
CAPC1005X5	10nF	CC0402_10NF	C35, C53, C54, C55, C56, C69, C70, C71,	Non-polarised capacitor	9
CAPC1005X5	1uF	CC0402_1UF_1	C36, C37, C88, C89, C94, C99	Non-polarised capacitor	6
CAPC1709X9	4.7uF	CC0603_4.7UF	C39, C43, C44	Non-polarised capacitor	3
CAPC1608X8	2.2nF	CC0603_2.2NF	C49, C50, C51, C52, C65, C66, C67, C68	Non-polarised capacitor	8
CAPC1608X9	220nF	CC0603_220NF	C84	Non-polarised capacitor	1
CAPC3216X1	47uF	CC1206_47UF	C86	Non-polarised capacitor	1
FP-SMAJ-	SMAJ40CA	CMP-08607-	D1	TVS DIODE 40V 64.5V DO214AC	1
VISH-SMB-	VS-	CMP-2000-	D2	Schottky Rectifier, 1.0 A, 100 V, -55 to 175	1
DIOD-B5X0C-	B560C-13-F	CMP-2000-	D3	Schottky Barrier Rectifier, 5 A, 60 V, -55 to	1
FP-SOT-666-	USBLC6-2P6	CMP-12225-	D4, D5, D7, D8	TVS DIODE 5.25V 17V SOT666	4
DIOD-	1N4148W-7-F	CMP-2000-	D6	Fast Switching Diode, 100 V, 0.15 A, -65 to	1
FP-SOD123-	1N5819HW-7-F	CMP-0303-	D9, D10, D11, D12, D21, D22, D23, D24	DIODE SCHOTTKY 40V 1A SOD123	8
SOT23_BAV9	SBAV99LT1G	BAV99	D13, D14, D15, D16, D25, D26, D27, D28	Dual Series Switching Diodes, 3-Pin SOT-23,	8
VISH-DO-	SMAJ33CA-	CMP-0548-	D17, D18, D19, D20, D29, D30, D31, D32	Transient Voltage Suppressor, 6.40 to 231 V,	8
FP-MMZ1608-	MMZ1608Q221B	CMP-08257-	FB1, FB2, FB3, FB5, FB7	Signal Line Ferrite Bead 220Ω @ 100MHz	5
FP-BLM15-	BLM15BD102SN	CMP-0686-	FB4	Chip Ferrite Bead, 0402, 1000Ω @ 100MHz,	1
FP-BLM18-	BLM18PG121SN	CMP-0686-	FB6	Chip Ferrite Bead, 0603, 120Ω @ 100MHz,	1
15EDGVC-	15EDGVC-3.5-	15EDGVC-3.5-	J1	15EDGVC-3.5-16P, розетка под клеммник	1
50A-20LB,	0.5mm, 20pin,	50A-20LB	J2	50A-20LB, ZIF разъем FFC/FPC SMT 0.5mm	1
15EDGVC-	15EDGVC-3.5-	15EDGVC-3.5-	J3	15EDGVC-3.5-08P, розетка под клеммник 8	1
HY951180A	HY951180A	HY951180A	J4, J5	1x1 Port Tab upright RJ45 Connector with	2
FP-TSW-104-	TSW-104-23-T-S	CMP-02766-	JH1	CONN HEADER VERT 4POS 2.54MM	1
FP-	LQH58PN100M	CMP-06042-	L1	Wire Wound Ferrite Inductor for Power Lines	1
B82472G610	SPRH74-150M	SPRH74-150M	L2	Inductor SMD	1
FP-	VLS3015CX-	VLS3015CX-	L3	10uH, 1.3A	1
XTAL_EPSO	25MHz	XTAL_25MHZ_E	Q1, Q2	Crystal 4 pins 1=XTAL Control	2
ONSC-SOT-	BC858CLT1G	CMP-1048-	Q3, Q4, Q5, Q6, Q11, Q12, Q13, Q14	General Purpose Transistor, PNP Silicon, 3-	8
FP-S-03946-	TP0610K-1T-E3	CMP-02550-	Q7, Q8, Q9, Q10, Q15, Q16, Q17, Q18	MOSFET P-CH 60V 185MA SOT23-3	8
FP-SOT23-	AO3400A	CMP-83955-	Q19	MOSFET N-CH 30V 5.7A SOT23	1
RESC1005X4	12k1	R0402_12K1_1	R1	Resistor - 1%	1
RESC1005X4	180k	R0402_180K_1	R2	Resistor - 1%	1
RESC1005X3	31k6	R0402_31K6_0	R3	Resistor - 0.1%	1
RESC1005X3	10k	R0402_10K_0.1	R4, R12, R31, R32, R33, R34, R35, R36,	Resistor - 0.1%, Resistor - 1%	14
RESC1005X4	49R9	R0402_49R9_1	R5, R6, R7, R8, R23, R24, R26, R28	Resistor - 1%	8
RESC1005X4	1k	R0402_1K_1%	R9, R10, R14, R25, R27, R29, R30	Resistor - 1%	7
RESC3216X6	1M	R1206_1M_1%	R11	Resistor - 1%	1
RESC1005X4	4k7	R0402_4K7_1%	R13	Resistor - 1%	1
RESC1005X4	5k1	R0402_5K1_1%	R15	Resistor - 1%	1
FP-	CRCW040211K	CMP-02407-	R16	RES Thick Film, 11.8kΩ, 1%, 0.063W,	1
RESC1005X4	1k5	R0402_1K5_1%	R17, R18	Resistor - 1%	2
RESC1005X4	22	R0402_22R_1%	R19, R20, R21, R22	Resistor - 1%	4
RESC1608X5	1k1	R0603_1K1_0.1	R37, R38, R39, R40, R53, R54, R55, R56	Resistor - 0.1%	8
RESC1608X5	15	R0603_15R_1%	R41, R42, R43, R44, R45, R46, R47, R48,	Resistor - 1%	16
RESC1608X5	82k	R0603_82K_1%	R65	Resistor - 1%	1
RESC1608X5	22k	R0603_22K_1%	R66	Resistor - 1%	1
RESC1608X5	2k2	R0603_2K2_1%	R68	Resistor - 1%	1
RESC1608X5	33	R0603_33R_1%	R69, R71, R72	Resistor - 1%	3
BOUR-CAY10	CAY10-330J4LF	CAY10-330J4LF	RA1, RA2	Chip Resistor Array, 33 Ohm, +/- 5%, 62.5	2
FP-	TPS54360BQDD	CMP-04918-	U1	IC REG BCK SPLIT RAIL ADJ 8SOPWR	1
FP-LQFP7x7-	KSZ8863RLL	CMP-06281-	U2	IC ETHERNET SW 3PORT 48-LQFP	1
FP-LQFP100-	STM32F407VGT	CMP-0243-	U3		1
FP-	XTR111AIDGQR	CMP-04883-	U4, U5, U6, U7, U8, U9, U10, U11	IC PREC VOLT-CURR CONV/TX 10MSOP	8
FP-	TPS560430YFQ	CMP-04918-	U12	IC POWER MANAGEMENT	1
DGN0008C-	TPS7A1633DGN	CMP-0394-	U13	Single Output LDO, 100 mA, Fixed 3.3 V	1
URB2424S-	URB2424S-	URB2424S-	U14	DC/DC изолированный 1600 VDC/s	1
SOP64P600X	CA-IS3741LB	CA-IS3741LB	U15	Драйвер-изолятор данных CA-IS3741LB, 4	1
DAC80508ZR	DAC80508ZRTE	DAC80508ZRTE	U16	Octal, 16-Bit, SPI, Voltage Output DAC with	1
MCP23S17-S	MCP23S17T-	MCP23S17T-	U17	16-Bit I/O Expander with Serial Interface, 28-	1
CA-	CA-IS3722LS	CA-IS3722LS	U18	Драйвер-изолятор данных CA-IS3722LG, 2	1
Approved					254
Notes					